

Comparator

→ Comparator is a combinational logic circuit that compares the magnitudes of two binary quantities to determine which one number has less than, equal or greater magnitude.

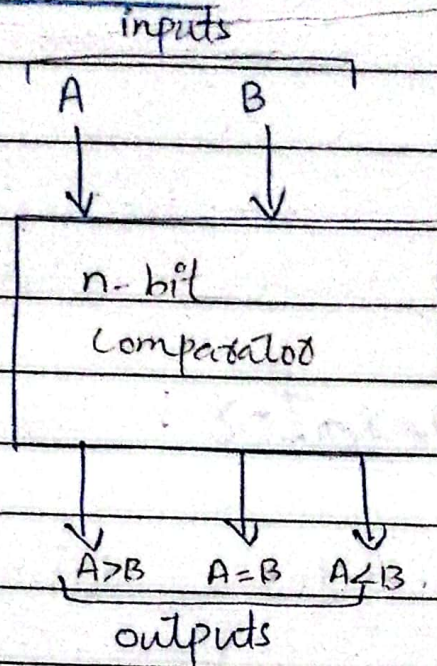
→ In other words, a comparator determines the relationship of two binary numbers/quantities.

→ A exclusive-NOR gate can be used as a basic comparator.

→ Three binary variables are used to indicate the outcomes of the comparison.

$A > B$, $A < B$, $A = B$

Block Diagram of n-bit Comparator:-



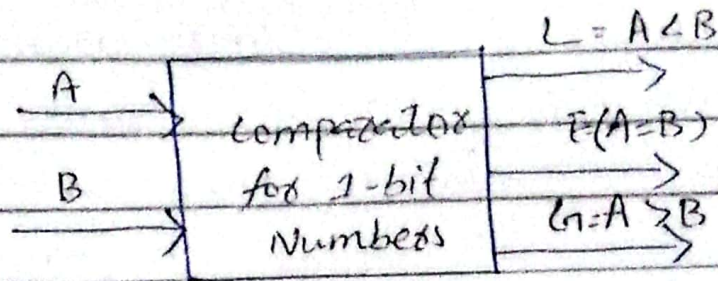
Comparators

- 1-Bit Comparator
- 2-Bit Comparator
- 4-Bit Comparator

1-Bit Magnitude Comparator:

- A comparator used to compare two numbers each of single bit is called single bit comparator.

Block Diagram (Single Bit Comparator) :-

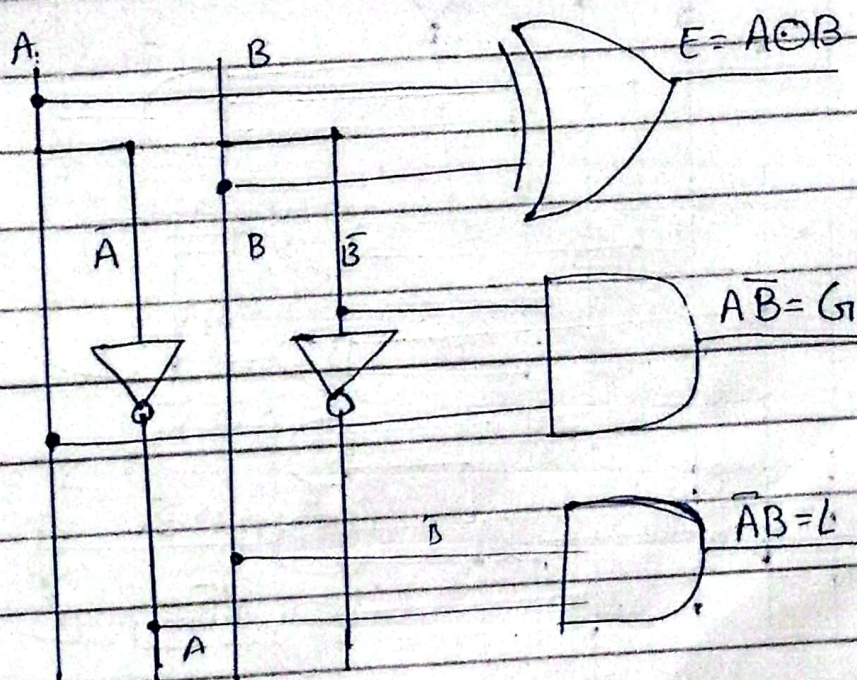


A	B	G $A > B$	E $A = B$	L $A < B$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

$$G = A\bar{B}$$

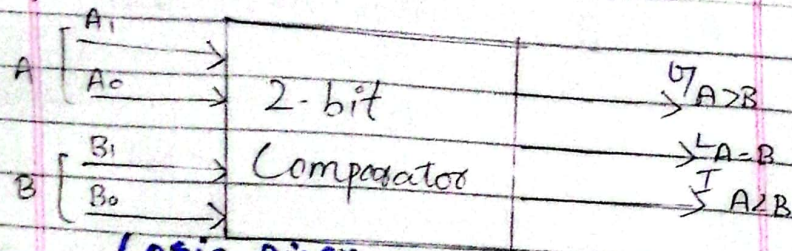
$$E = \bar{A}\bar{B} + AB = \overline{A \oplus B} = A \odot B$$

$$L = \bar{A}B$$

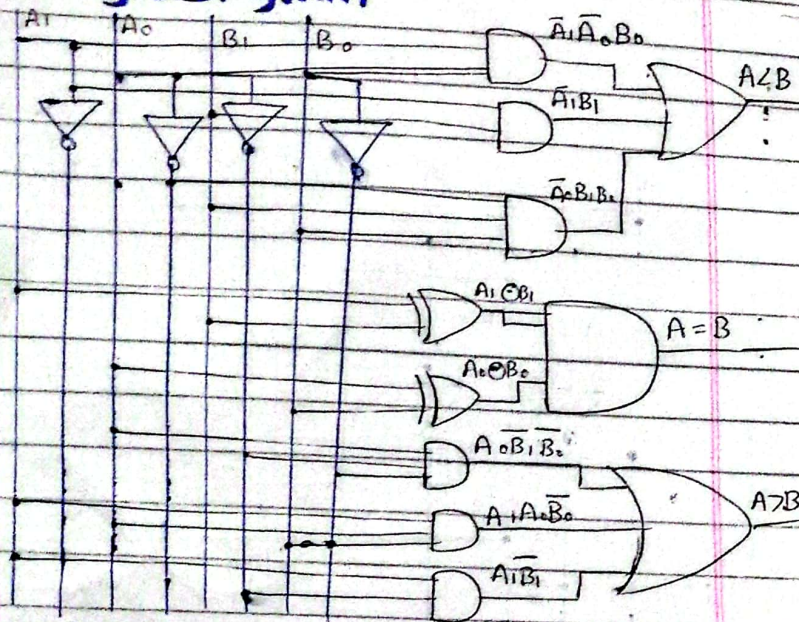


2-Bit Comparators

→ A 2-bit comparator compares two binary numbers, each of two bits and produces their relation such as one number is equal or greater than or less than the other.



Logic Diagram



A ₁	A ₀	B ₁	B ₀	L = A < B	E = A = B	G = A > B
0	0	0	0	0	1	0
0	0	0	1	0	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	0	0	0
1	0	0	0	1	0	0
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	0	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	1	0
1	1	1	1	0	1	0

$$G = \bar{A}_1 A_0 \bar{B}_1 \bar{B}_0 + A_1 \bar{A}_0 \bar{B}_1 \bar{B}_0$$

A ₁ A ₀	B ₁ B ₀	01	11	10
00				
01		1		
11		1	1	1
10		1	1	

$$G = A_1 \bar{B}_1 + \bar{B}_1 \bar{B}_0 A_0 + A_1 A_0 \bar{B}_0$$

$A_1 A_0$ \ $B_1 B_0$	00	01	11	10
00		1	1	1
01			1	1
11				
10			1	

$$L = \bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0$$

$A_1 A_0$ \ $B_1 B_0$	00	01	11	10
00	1			
01		1		
11			1	
10				1

$$= \bar{A}_1 \bar{A}_0 \bar{B}_1 \bar{B}_0 + \bar{A}_1 \bar{A}_0 \bar{B}_1 B_0 + A_1 \bar{A}_0 B_1 \bar{B}_0 + A_1 \bar{A}_0 B_1 B_0$$

$$= \bar{A}_1 \bar{B}_1 (\bar{A}_0 \bar{B}_0 + A_0 B_0) + A_1 B_1$$

$$= (\bar{A}_1 \bar{B}_1 + A_1 B_1) (\bar{A}_0 \bar{B}_0 + A_0 B_0)$$

$$= (A_1 \odot B_1) (A_0 \odot B_0)$$